



Cellular mechanosensation controlling real-time robot decisions.

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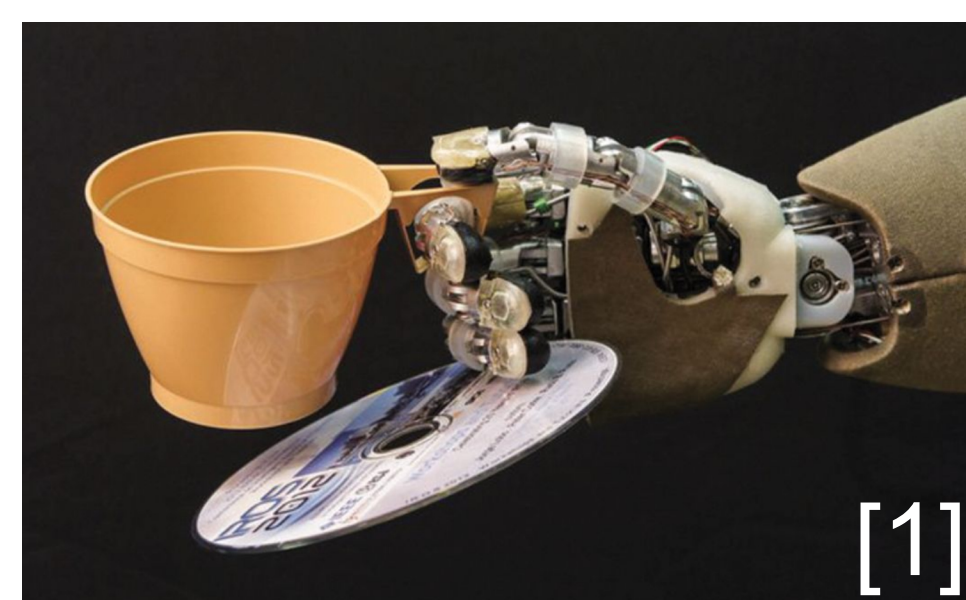


Abstract

Living systems convert mechanical cues into decisions with speed and sensitivity that remain difficult to reproduce in robots. Here we report a biohybrid mechanosensor that uses mechanically responsive bioluminescent algae as an onboard feedback element for robotic decision making. The cells are enclosed in a soft hemispherical blister network that transforms external deformation into localized cellular light emission which are converted into real-time machine-readable outputs. The device geometry and excitation rate enable the sensor to distinguish mechanical events during loading and release. Integrated with a robotic gripper, the system directly controls manipulation decisions: arresting grip closure, detecting slip after grasp, and recognizing unplanned contact. This work establishes a route toward biohybrid sensors that couple evolved biological responsiveness to autonomous machines for decentralized decision making.

Introduction

Mechanosensation allows organisms to **convert force and deformation** into diverse behavior and can expand robotic **closed-loop control** across many modes of manipulation [1, 2].

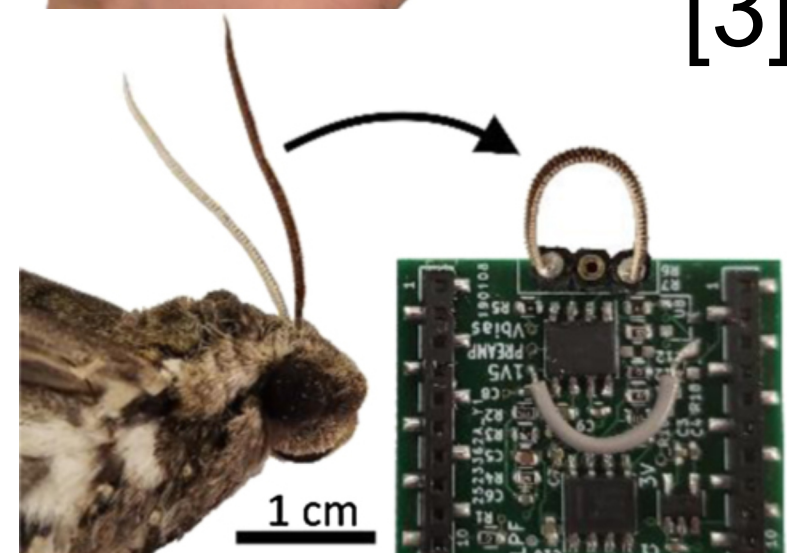


[1]

Biohybrid systems have **outperformed** man-made sensors in modalities like **olfaction**, showcasing the potential benefits of tapping into sensing pathways **honed by evolution** [3, 4].

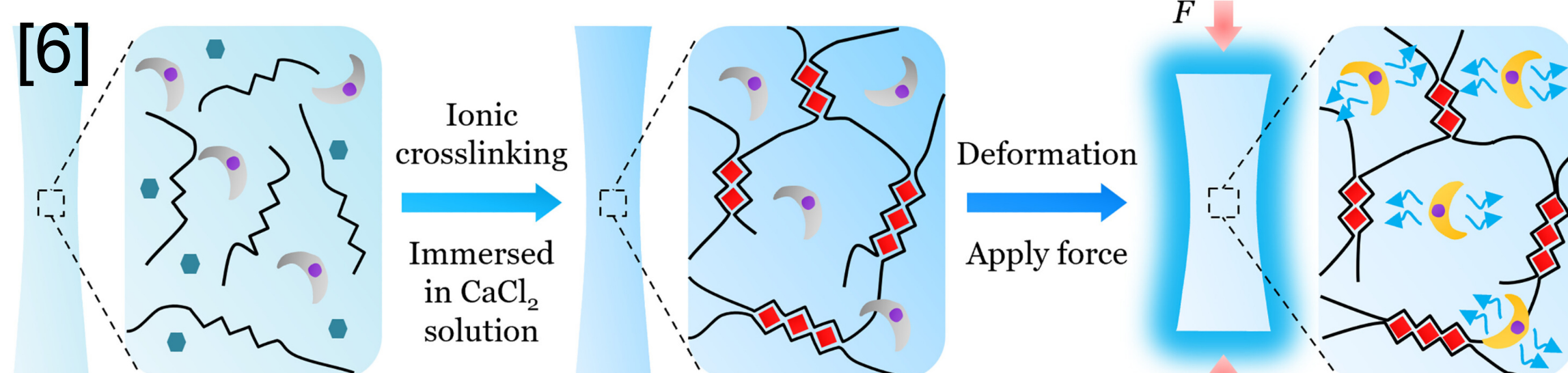


[3]



1 cm

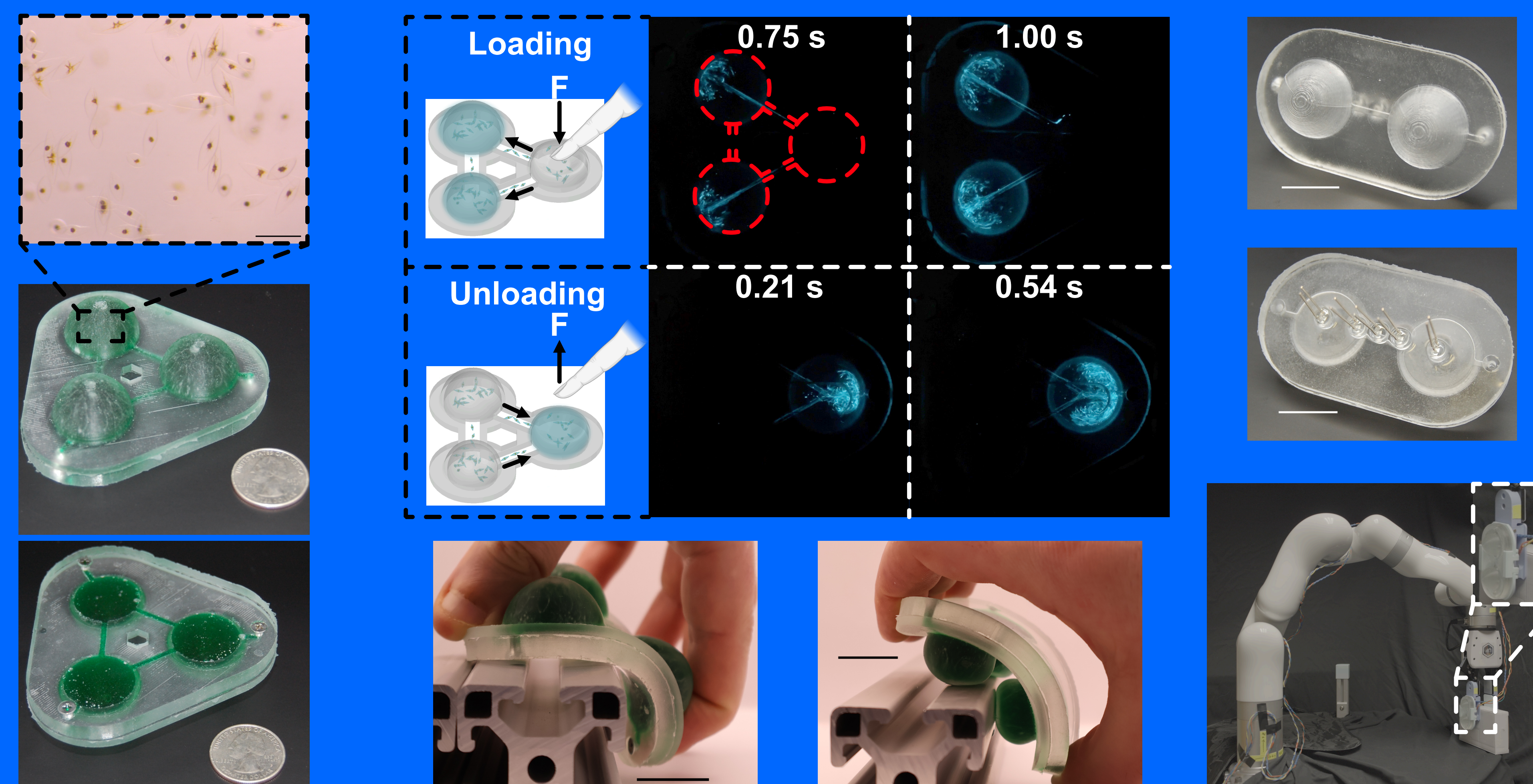
Existing biological systems like **slime mold** and **algae** have demonstrated measurable **sensitivity to mechanical stimuli** [5, 6].



We are exploring **bioluminescent algae** as a **biohybrid mechanosensor** for **real-time feedback** due to its robust culture protocol, signal strength, and ~10 ms onset [7]. Our soft sensor converts deformation to internal fluidic shear, triggering algae response and transducing it for our robotic arm via onboard optoelectronics.

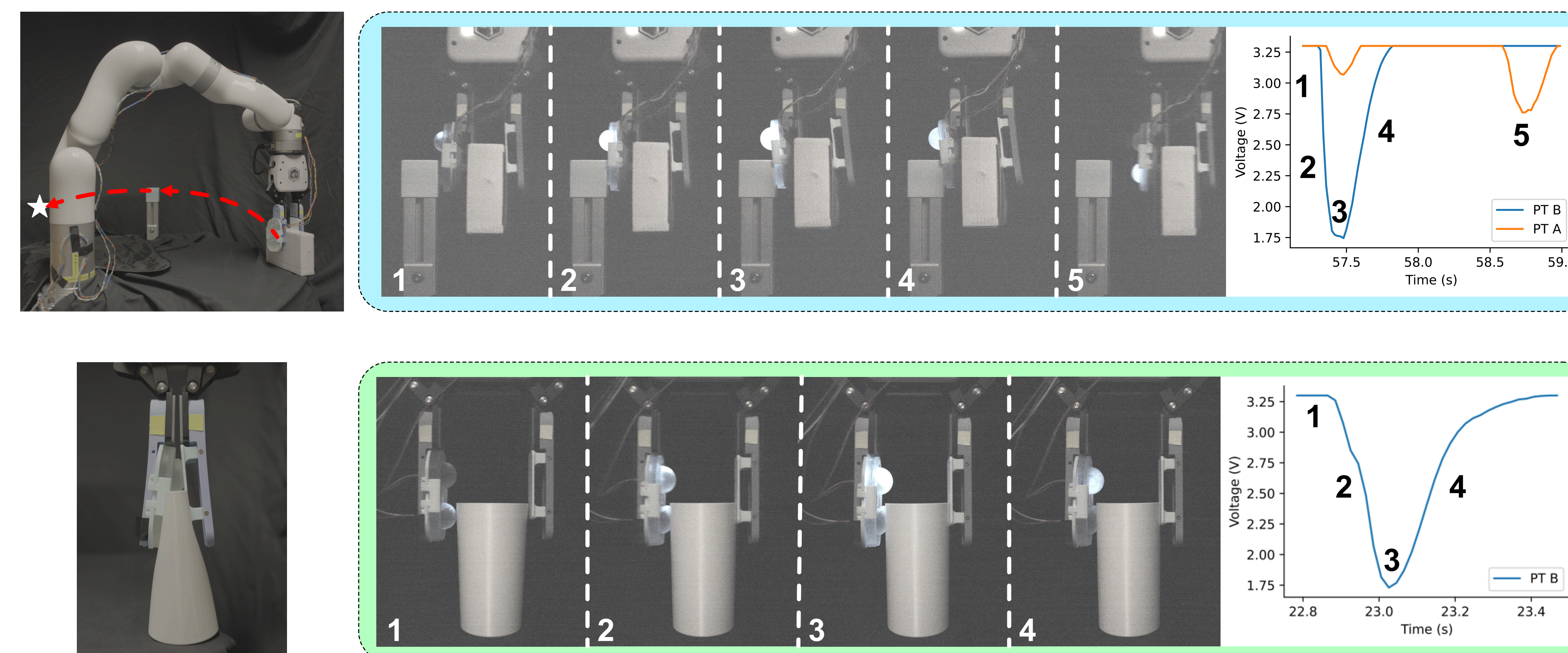
Project Description

Soft, reusable biohybrid mechanosensor employs living **bioluminescent algae** as the sensing interface for **robot decision making**. Contact evokes cellular response for onboard sensing, inducing robotic state transitions during manipulation.



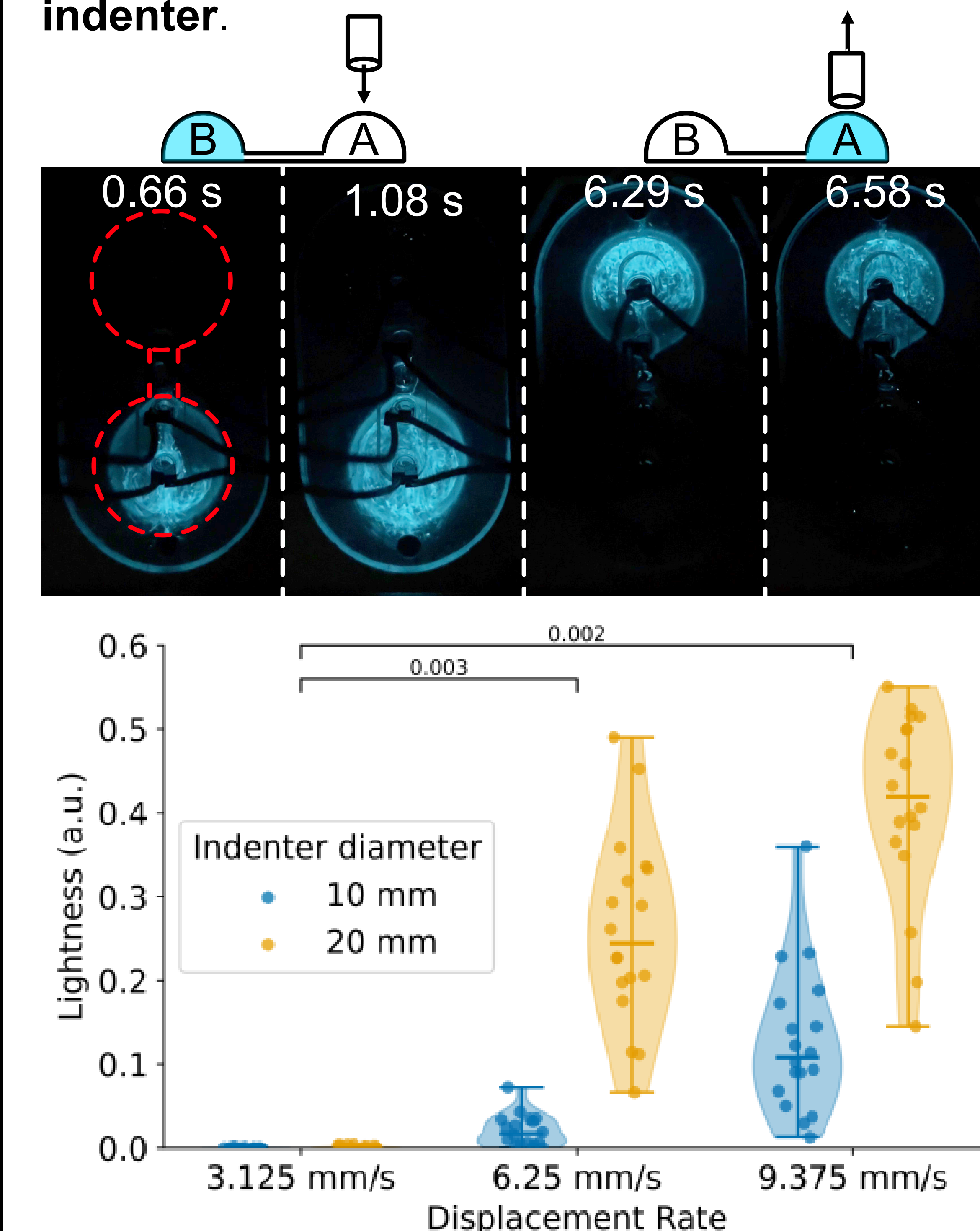
Integrated Mechanosensor Results

Biohybrid mechanosensor provides **real-time feedback** to robot, including both **obstacle detection** and **grasp arrest**.



Luminescence Results

Sensor **luminescence** is influenced both by the **incident rate** of contact and the **size of indenter**.



Conclusions

- Designed **soft, reusable biohybrid mechanosensor** with bioluminescent algae
- Explored **luminescent response** to **various indentations**
- Real-time biological feedback** enabled to **autonomous pick-and-place sequence**

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[4] Y. Hirata, et al., *Lab Chip* (2021)

[5] A. Adamatzky, *IEEE Sensors Journal* (2014)

[6] C. Li, et al., *Science Advances* (2023)

[7] M. I. Latz, et al., *Journal of Experimental Biology* (2008)

LeDuc Lab



SHRED Lab

